

Zurich ServiceNow AI Platform Capabilities

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Service Graph Connector for SolarWinds

Use the Service Graph Connector for SolarWinds to pull in data from the SolarWinds software into your ServiceNow instance.

The Service Graph Connector for SolarWinds pulls in asset inventory data (hardware and software) from the SolarWinds database into the ServiceNow® Configuration Management Database (CMDB) application.

Request apps on the Store

Visit the [ServiceNow Store](#) website to view all the available apps and for information about submitting requests to the store. For cumulative release notes information for all released apps, see the [ServiceNow Store version history release notes](#).

Supported versions

- Supported SolarWinds Orion minimum versions:
 - 2019.4 HF6
 - 2020.2.1. HF2
- Supported ServiceNow versions:
 - Washington DC
 - Xanadu
 - Yokohama

Use cases

The following are examples on how you can use the Service Graph Connector:

- Automatic normalization of asset information for hardware, virtualization and cloud resources, and software.
- Ability to configure and save synchronization schedules.

Guided setup

The guided setup for the Service Graph Connector for SolarWinds provides an organized sequence of tasks to configure the integration on your instance. To access the guided setup, see [Configure Service Graph Connector for SolarWinds](#).

CMDB integrations dashboard

The Integration Commons for CMDB store app provides a dashboard with a central view of the status, processing results, and processing errors of all installed Service Graph Connectors. You can see metrics for all integration runs. You can filter the view to a specific integration, a specific time duration, or a specific integration run. For more details about monitoring SolarWinds integrations in the CMDB Integrations Dashboard, see [Using the CMDB Integrations Dashboard](#).

Data mapping

Data from data sources in the SolarWinds software is mapped and transformed into ServiceNow CMDB tables using the Robust Transform Engine (RTE). Data is inserted into ServiceNow CMDB using the Identification and Reconciliation Engine (IRE).

When you complete setting up the connection, you can configure the integration to periodically pull data from the SolarWinds software.

The data is loaded into staging tables and then inserted into the following CMDB target tables:

- AIX Server [cmdb_ci_aix_server]
- Availability Zone [cmdb_ci_availability_zone]
- CI Relationship [cmdb_rel_ci]
- Cloud Network [cmdb_ci_network]
- Cloud Service Account [cmdb_ci_cloud_service_account]
- Cloud Subnet [cmdb_ci_cloud_subnet]
- Computer [cmdb_ci_computer]
- Disk [cmdb_ci_disk]

- Hardware [cmdb_ci_hardware]
- Hardware Type [cmdb_ci_compute_template]
- Hyper-V Server [cmdb_ci_hyper_v_server]
- IIS Virtual Directory [cmdb_ci_iisdirectory]
- Image [cmdb_ci_os_template]
- IP Address [cmdb_ci_ip_address]
- Linux Server [cmdb_ci_linux_server]
- Logical Datacenter [cmdb_ci_logical_datacenter]
- Microsoft iis Web Server [cmdb_ci_microsoft_iis_web_server]
- MS SQL Server [cmdb_ci_db_mssql_server]
- MS SQL DataBase [cmdb_ci_db_mssql_database]
- Network Adapter [cmdb_ci_network_adapter]
- Network Gear [cmdb_ci_netgear]
- Serial Number [cmdb_serial_number]
- Server [cmdb_ci_server]
- Software Installation [cmdb_sam_sw_install] (If the Software Asset Management (SAM) application is installed.)
- Software Instance [cmdb_software_instance] (If the SAM application is not installed.)
- Software [cmdb_ci_spkg] (If the SAM application is not installed.)
- Solaris Server [cmdb_ci_solaris_server]
- Storage Volume [cmdb_ci_storage_volume]
- UNIX Server [cmdb_ci_unix_server]
- VM Instance [cmdb_ci_vm_instance]
- Windows Server [cmdb_ci_win_server]

For more information on where data is saved when pulling data from SolarWinds, see [CMDB classes targeted in Service Graph Connector for SolarWinds](#).

You can use the IntegrationHub ETL app to view the data maps. See [IntegrationHub ETL](#) for more information.

Related concepts

- [Service Graph Connectors](#)

Configure Service Graph Connector for SolarWinds using SGC Central

Use the playbook available with the SGC Central application to set up the Service Graph Connector for SolarWinds for pulling in SolarWinds data into the CMDB.

Before you begin

Install Service Graph Connector for SolarWinds version 2.6.0 or later from the ServiceNow Store. For ServiceNow Store installation steps, see [Install a ServiceNow Store application](#).

Air gap connections are available from version 2.5.0 of the Service Graph Connector for SolarWinds. You can set up either standard data sources or air gap data sources. Before you set up air gap data sources, you must configure the air gap solution for the Service Graph Connector for SolarWinds in your high-secure server and low-secure server. See [Configuring air gap connections for Service Graph Connector for SolarWinds](#).

Role required: SGC-Admin (sn_cmdb_int_util.sgc_admin)

Note: The admin user role is required to run background scripts and to provide table-level access to the SGC-Admin user.

About this task

The playbook experience for onboarding connectors is activated with SGC Central in the CMDB Workspace. To configure the SGC Central

application, see [Configuring SGC Central](#) and for more information on how to interact with a playbook, see [Interact with Playbook](#).

Procedure

1. Navigate to **Workspaces > CMDB Workspace**.
2. In the CMDB Workspace, select **SGC Central**.
3. On the Dashboard page, select **Create connection**.

Tip: Alternatively, you can select **Create connection** on the All connections page.
4. On the Create connection window, select the SolarWinds connector type, and then select **Create connection**.
5. Complete the initial prerequisites when setting up a connection for the first time using a connector.

Note: This step is required only during the first-time setup. See [Perform initial setup tasks when creating a connection in SGC Central](#).

6. Complete the setup for configuring the connector for importing data.
 - a. Select the data source type.
 - a. In the **Setup** stage of the playbook, select the **Select data source type** activity.
 - Select **Standard data sources** if you're working with environments that have direct internet access.

Proceed to step [6.c](#).

 - Select **Air gap data sources** if you're working with restricted environments that don't have direct internet access.
 - b. Select **Continue**.
 - b. Download and run the PowerShell scripts to set up an air gap connection.

Note: Before you add an air gap data source, you must configure the air gap solution for the Service Graph Connector for SolarWinds in your high-secure and low-secure servers. See [Configuring air gap connections for Service Graph Connector for SolarWinds](#).

- a. In the **Setup** stage of the playbook, select the **Download PowerShell scripts** activity.

The **Download PowerShell scripts** activity appears only when you select **Air gap data sources** as the data source type in step 6.a.

- b. Select **Download PowerShell scripts**.

- c. Run the PowerShell scripts.

For more information on configuring an air gap connection, see [Configuring air gap connections for Service Graph Connector for SolarWinds](#).

- d. After configuring the air gap connection, select **Continue**.

- c. Create and test a connection.

- a. In the **Setup** stage of the playbook, select the **Create and test connection** activity.

- b. On the form, fill in the fields.

- For standard data sources:

Create and test connection form

Field	Description
Connection Name	Display name for the connection.
Connection URL	Connection host name for SolarWinds.

Field	Description
User name	User name for SolarWinds authentication.
Password	Password for SolarWinds authentication.
Use MID server	Option to use a MID Server. Select this check box.
MID selection	Select a MID Server from the list. This field appears only when the Use MID Server check box is selected.
MID application	Select a MID Server application from the list. This field appears only when the Use MID Server check box is selected.

- For air gap data sources:

Create and test connection form

Field	Description
Connection name	Display name for the connection.
Parent directory	Path of the SolarWinds directory where the data is stored in your low-secure server.
Archive data after retrieval	Option to archive data after retrieval.

Field	Description
Archive path [Optional]	Folder where the data is to be archived if the Archive data after retrieval check box is selected.
MID server	Name of the MID Server that you deployed in your low-secure server. See Configure air gap solution for Service Graph Connector for SolarWinds in a low-secure server .

- c. Select **Create and test connection**.
- d. Once the connection test is complete, select **Continue**.
- d. Configure the properties of the connector to access resources.
 - a. In the **Setup** stage of the playbook, select the **Set configuration properties** activity.
 - Select **SAM installed** if a Service & Application Monitor (SAM) module is installed on the SolarWinds instance.
 - Select **NPM installed** if a Network Performance Monitor (NPM) module is installed on the SolarWinds instance.
 - b. Select **Continue**.
- e. Configure the import schedule to import data at regular intervals.
 - a. In the **Setup** stage of the playbook, select the **Configure import schedule** activity.
 - b. Expand the Parent scheduled data import within the Import schedules list, and select the SG-Solarwinds Hardware import schedule.

- c. In the Configure import schedule dialog box, select the **Active** check box, and then fill in the run schedule and time details.

For more information, see [Schedule a data import](#).

- d. Select **Save**.

Alternatively, select **Execute Now** to execute the import schedule immediately.

- e. Select **Continue**.

- f. In the **Setup** stage of the playbook, select the **Confirm connection creation** activity to verify whether the connection was created.

What to do next

Select **View all connections** to review the connection details. The configured connection appears in the Installed connections list.

Related concepts

- [Service Graph Connector for SolarWinds](#)
- [Accessing the connection details of Service Graph Connector for SolarWinds](#)
- [Configuring air gap connections for Service Graph Connector for SolarWinds](#)

Related reference

- [CMDB classes targeted in Service Graph Connector for SolarWinds](#)

Configure Service Graph Connector for SolarWinds

Set up authentication credentials and scheduled jobs to import SolarWinds data into your CMDB.

Before you begin

Important: Unless there are configuration issues, use the SGC Central view in the CMDB Workspace to configure the connection for the connector, as the guided setup method is being deprecated.

To use this Service Graph Connector, you need a subscription to a Subscription Unit that is based in the IT Operations Management (ITOM) Visibility application or in the ITOM Discovery application. As defined in the section titled "Managed IT Resource Types" in [ServiceNow Subscription Unit Overview](#) for your subscription, for managed IT resources that are created or modified in the CMDB by this Service Graph Connector, but that aren't yet managed by [ITOM Visibility](#) or [ITOM Discovery](#), these resources will increase Subscription Unit consumption from that application. Review your current Subscription Unit consumption within ITOM Visibility or ITOM Discovery to ensure available capacity.

Before you start the configuration, navigate to **System Definition > Business Rules**, and deactivate the ValidateServiceAccountID business rule.

Air gap connections are available from version 2.5.0 of the Service Graph Connector for SolarWinds. You can set up either regular data sources or air gap data sources. Before you set up air gap data sources, you must configure the air gap solution for the Service Graph Connector for SolarWinds in your high-secure server and low-secure server. See [Configuring air gap connections for Service Graph Connector for SolarWinds](#).

Dependencies and requirements:

- The [Integration Commons for CMDB](#) store app, which is automatically installed.
- The CMDB CI class models store app, which is automatically installed. See [CMDB CI Class Models](#).
- Discovery Core plugin (com.snc.discovery.core), which is automatically installed by Discovery.
- ITOM Discovery License plugin (com.snc.itom.discovery.license). You must activate this plugin.

- ITOM Licensing plugin (com.snc.itom.license). For more information, see [Request Discovery](#).
- SolarWinds Orion Platform.
- SolarWinds Server & Application Monitor and/or Network Performance Monitor.

Roles required:

- To configure the ServiceNow platform: SGC-Admin (sn_cmdb_int_util.sgc_admin)

Note: The admin user role is required to run background scripts and to provide table-level access to the SGC-Admin user.
- To access SolarWinds data: User with SWIS (SolarWinds Information Service) access (the same as through the Orion website, not database users).

About this task

Note: SolarWinds Integration can work with or without Service & Application Monitor (SAM) and Network Performance Monitor (NPM) being installed.

Procedure

1. Navigate to **All > Service Graph Connectors > SolarWinds > Setup**.
2. On the Getting started page, select **Get Started**.
3. Configure your authentication credentials used to connect to the SolarWinds SWIS API.
 - a. On the Service Graph Connector for SolarWinds page, in the Configure the connection section, select the **Configure the authentication credentials used to connect to SolarWinds SWIS API** task.
 - b. For the Configure the authentication credentials used to connect to SolarWinds SWIS API task, select **Configure**.

- c. Fill in the fields on the Basic Auth Credentials form.

Basic Auth Credentials form

Field	Description
Name	Human-readable name for this credential. This field is automatically set. You can optionally change this setting.
User name	SolarWinds user name. Note: The SolarWinds user must have a role with read privileges to the SolarWinds Information Service API for all SolarWinds data sources that the integration can pull data.
Password	SolarWinds password that is stored in the database in encrypted form.
Active	Option to check whether this credential is active.
Credential alias	Advanced selection criteria for this credential.
Order	Order in which credentials are tried. Smaller numbers are tried first.

- d. Select **Update** if required.
- e. Set the Configure the authentication credentials used to connect to the SolarWinds SWIS API task to complete by selecting **Mark as Complete**.
4. Configure the SolarWinds HTTP connection.

- a. For the Configure SolarWinds HTTP connection task, select **Configure**.
- b. Review the HTTP(s) Connection form, and fill in the fields as required.

HTTP(s) Connection form

Field	Description
Name	Name of the connection. This field is automatically set. You can optionally change the name.
Use MID server	Option to enable the connection to use a MID server.
Host	Target host value used by the connection. This field is automatically set by the connection URL.
Credential	Credential value used by the connection.
Connection alias	Connection alias value with which the connection can be referred.
URL builder	URL builder that is used to build the connection URL.
Mutual authentication	Option to enable mutual authentication.
Protocol	Underlying protocol used by the connection.
Active	Option to activate the HTTP connection.

Field	Description
Domain	Domain to which the connection belongs.
Override default port	Target port value used by the connection.
Base path	Base path for HTTP(S) connection that is required but should not be modified.

- c. Select **Update**, if required.
 - d. Set the Configure SolarWinds HTTP connection task to complete by selecting **Mark as Complete**.
5. Configure the SolarWinds modules.
- a. For the Configure SolarWinds Modules task, select **Configure**.
 - b. Update the **Value** for the sn_solarwinds_inte.npm_installed and the sn_solarwinds_inte.sam_installed properties in one of the following ways:
 - If the respective module is installed on the SolarWinds instance, then set the fields to **true**.
 - If the respective module is not installed on the SolarWinds instance, then set the fields to **false**.

Note: This update changes the API class and the data that is returned from them.
 - c. Set the Configure SolarWinds Modules task to complete by selecting **Mark as Complete**.
6. Test the connection.
- a. For the Test the connection task, select **Configure**.
 - b. Review the fields on the Data Source form, which are automatically set.

Data Source form

Field	Description
Name	Unique name for this data source.
Import set table label	Label of the table that will be created for this data source.
Import set table name	Name of the table that will be created for this data source.
Type	Data storage type of the data to be imported.
Data in single column	Data in single column.
Application	Application containing this record.
Data stream action	The Data Source request action that will be invoked to get data.

- c. Test the connection by selecting the **Test Load 20 Records** related link.

Testing the connection takes a few moments, after which the page refreshes to show the test results.

This step tests the SG-Solarwinds Hardware data source, and verifies that data is loaded into the staging table. A successful connection for the SG-Solarwinds Hardware data source means that all SolarWinds data sources connect successfully, so you do not need to individually test all data sources.

The connection is successful if the **HTTP Status** is **200**. If there is an **Error Code** and **Error Message**, the connection failed and further troubleshooting is required.

- d. Navigate back to the guided setup, and set the Test the connection task to complete by selecting **Mark as Complete**.

7. Add multiple instances.

Note: If you do not need to add multiple instances, you can skip this step.

- a. On the Service Graph Connector for SolarWinds page, in the Add Multiple Instances section, select the **Update Data Source Access** task.
- b. For the Update Data Source Access task, select **Configure**.
- c. Select the Data Source [sys_data_source] table.
- d. Select the **Global** application scope by using the application picker.
- e. In the Application Access related list, select the **Can create, Can update, and Can delete** check boxes, if not already selected.
- f. Select **Update** to close the tab and return to the guided setup.
- g. Modify the application scope again to **Service Graph Connector for SolarWinds** by using the application picker.
- h. Set the **Update Data Source Access** task to complete by selecting **Mark as Complete**.
- i. Repeat the steps 7.d to 7.g for the Update Scheduled Data Import Access task with the Scheduled data import [scheduled_data_set] table and for the Update Value Access task with the Value [sys_variable_table] table, and set the tasks to complete by selecting **Mark as Complete**.

8. Clear the cache for the new connection.

- a. Select the **Clear Cache for Datasource and Import set** task, and then select **Configure**.
- b. Clear the cache by selecting **Global** from the Scope menu.
- c. Enter the following script.

```
GlideTableManager.invalidateTable("sys_data_source");
GlideCacheManager.flushTable("sys_data_source")
```

```
);

    GlideTableManager.invalidateTable("scheduled_
import_set");
    GlideCacheManager.flushTable("scheduled_import
_set");

    GlideTableManager.invalidateTable("sys_variab
le_value");
    GlideCacheManager.flushTable("sys_variable_val
ue");

    GlideTableManager.invalidateTable("sys_db_obj
ect");
    GlideCacheManager.flushTable("sys_db_object");
```

- d. Select **Run Script**.
 - e. From the Scope menu, select **Service Graph Connector for SolarWinds**.
 - f. Set the Clear Cache for Datasource and Import set task to complete by selecting **Mark as Complete**.
9. Add a connection to another SolarWinds instance.
- Note:** Confirm that the current scope is **Service Graph Connector for SolarWinds**.
- a. Select the **Add Another Connection** task, and then select **Configure**.
 - b. In the Select Data Source Type section, select either regular data sources or air gap data sources.
 - To add or edit regular data sources, select **Regular Data Sources**.
 - To add or edit air gap data sources, select **Airgap Data Sources**.

Note: Before you add an air gap data source, you must configure the air gap solution for the Service Graph Connector for SolarWinds in your high-secure and low-secure servers. See [Configuring air gap connections for Service Graph Connector for SolarWinds](#).

c. Either create or edit a connection.

- To create a new connection, select **Add Connection**.
- To edit an existing connection, select **Edit**.

d. Fill in or edit the fields on the form, as required.

- For regular data sources:

Create Connection form

Field	Description
Connection Name	Display name for the connection.
Connection URL	Connection host name for SolarWinds.
User name	User name for SolarWinds authentication.
Password	Password for SolarWinds authentication.

- For air gap data sources:

Create Connection form

Field	Description
Connection Name	Display name for the connection.
MID Server	Name of the MID server that you deployed in your low-secure server.

Field	Description
	See Configure air gap solution for Service Graph Connector for SolarWinds in a low-secure server.
Parent Directory	Path of the SolarWinds directory where the data is stored in your low-secure server.
Archive Data After Retrieval	Option to archive data after retrieval.
Archive Path [Optional]	Folder where the data is to be archived if the Archive Data After Retrieval check box is selected.

- e. Either add or save the connection.
 - To add a new connection, select **Create Connection**.
 - To save the edits for the existing connection, select **Edit Connection**.
- f. Navigate back to the guided setup, and select **Mark as Complete** to set the Add Another Connection task to complete.
- g. If required, set up the MID Server for the connection you created.
 - a. For the Configure Mid Servers task, select **Configure**.
 - b. Select the name of the connection you created.
 - c. Select the **Use MID server** check box.
 - d. Select **Update**.
 - e. Set the Configure Mid Servers task to complete by selecting **Mark as Complete**.

- h. If required, configure the connections in the Configure SolarWinds Modules section by selecting **Configure**.
- i. When you're finished, close the window, and navigate back to the guided setup.
- j. Set the Configure SolarWinds Modules task to complete by selecting **Mark as Complete**.
- k. For the Test New Connections task, select **Configure**.
 - a. Select the name of the data source associated with the newly created connection.
 - b. Select the **Test Load 20 Records** related link.

Note: If the displayed completion code is Success, then the sources are validated. But if the displayed completion code is Error, then there is an error that you must fix.
 - c. Navigate back to the guided setup, and set the Test New Connections task to complete by selecting **Mark as Complete**.

10. Set up scheduled import jobs.

- a. On the Service Graph Connector for SolarWinds page, in the Set up scheduled import jobs section, select the **Configure the scheduled import jobs** task.
- b. For the Configure the scheduled import jobs task, select **Configure**.
- c. Review the fields on the Scheduled Data Import form, which are automatically set.

Scheduled Data Import form

Field	Description
Name	Name of the scheduled job.
Data source	Data source record that defines the data to import.

Field	Description
Run as	Option to run the scheduled job with the credentials of the specified user.
Active	Option to activate the scheduled job. Select this option.
Concurrent Import	Function that loads the data from multiple import sets. The function then processes and transforms the data concurrently.
Partition Method	Partition method for the concurrent import set.
Partition Size	Import set size for early scheduling.
Execute pre-import script	Option to run a pre-import script before the import is performed.
Execute post-import script	Option to run a post-import script after the import is performed.
Application	Application containing this record.
Run	Frequency of running the import.
Conditional	Conditions under which this job is executed.

Note: All active SolarWinds scheduled jobs will run in their specified order after the SG-Solarwinds Hardware scheduled job runs. You can modify the **Active** setting for each SolarWinds scheduled job as appropriate for your integration.

- d. Select **Update** if required, and then select **Mark as Complete** to set the Set up scheduled import jobs task to complete.

Related tasks

- [Configure Service Graph Connector for SolarWinds using SGC Central](#)

Related concepts

- [Service Graph Connector for SolarWinds](#)
- [Accessing the connection details of Service Graph Connector for SolarWinds](#)
- [Configuring air gap connections for Service Graph Connector for SolarWinds](#)

Related reference

- [CMDB classes targeted in Service Graph Connector for SolarWinds](#)

CMDB classes targeted in Service Graph Connector for SolarWinds

When you complete setting up the connection, you can configure the integration to periodically pull data from Solarwinds. The data is saved in tables that extend from the Configuration item [cmdb_ci] table.

The following attributes in the Availability Zone [cmdb_ci_availability_zone] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id

Relationships created for Availability Zone

Parent class	Relationship type	Child class
Availability Zone [cmdb_ci_availability_zone]	Contains::Contained by	Cloud Subnet [cmdb_ci_cloud_subnet]
Availability Zone [cmdb_ci_availability_zone]	Contains::Contained by	Cloud Network [cmdb_ci_network]

The following attributes in the Cloud Key Pair [cmdb_ci_cloud_key_pair] table are populated by collected data:

Attribute label	Attribute name
Object ID	object_id
Name	name

Relationship created for Cloud Key Pair

Parent class	Relationship type	Child class
Cloud Key Pair [cmdb_ci_cloud_key_pair]	Hosted on::Hosts	Logical Datacenter [cmdb_ci_logical_datacenter]

The following attributes in the Cloud Network [cmdb_ci_network] table are populated by collected data:

Attribute label	Attribute name
Name	name

Attribute label	Attribute name
Object ID	object_id

Relationships created for Cloud Network

Parent class	Relationship type	Child class
Cloud Network [cmdb_ci_network]	Contains::Contained by	Cloud Subnet [cmdb_ci_cloud_subnet]
Cloud Network [cmdb_ci_network]	Hosted on::Hosts	Logical Datacenter [cmdb_ci_logical_datacenter]

The following attributes in the Cloud Service Account [cmdb_ci_cloud_service_account] table are populated by collected data:

Attribute label	Attribute name
Account Id	account_id
Name	name
Object ID	object_id
Datacenter Type	datacenter_type

The following attributes in the Cloud Subnet [cmdb_ci_cloud_subnet] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id

The following attributes in the Computer [cmdb_ci_computer] table are populated by collected data:

Attribute label	Attribute name
CPU core count	cpu_core_count
CPU core thread	cpu_core_thread
CPU name	cpu_name
CPU speed (MHz)	cpu_speed
CPU manufacturer	cpu_manufacturer
Is Virtual	virtual

Relationships created for Computer

Parent class	Relationship type	Child class
Computer [cmdb_ci_computer]	Virtualized by::Virtualizes	Virtual Machine Instance [cmdb_ci_vm_instance]
Computer [cmdb_ci_computer]	Contains::Contained by	Disk [cmdb_ci_disk]

The following attributes in the Disk [cmdb_ci_disk] table are populated by collected data:

Attribute label	Attribute name
Model ID	model_id
Device ID	device_id
Name	name
Disk space (GB)	disk_space
Free disk space (GB)	free_space

Attribute label	Attribute name
Size	size
Size bytes	size_bytes
Computer	computer
Manufacturer	manufacturer
Volume serial number	volume_serial_number

Relationship created for Disk

Parent class	Relationship type	Child class
Disk [cmdb_ci_disk]	Reference	Computer [cmdb_ci_computer]

The following attributes in the Hardware [cmdb_ci_hardware] table are populated by collected data:

Attribute label	Attribute name
Model ID	model_id
Name	name
DNS Domain	dns_domain
Manufacturer	manufacturer
Serial number	serial_number
Class	sys_class_name
Default Gateway	default_gateway
Fully qualified domain name	fqdn

Relationships created for Hardware

Parent class	Relationship type	Child class
Hardware [cmdb_ci_hardware]	Owns::Owned by	IP Address [cmdb_ci_ip_address]
Hardware [cmdb_ci_hardware]	Owns::Owned by	Network Adapter [cmdb_ci_network_adapter]

The following attributes in the Hardware Type [cmdb_ci_compute_template] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id

Relationship created for Hardware Type

Parent class	Relationship type	Child class
Hardware Type [cmdb_ci_compute_template]	Hosed on::Hosts	Logical Datacenter [cmdb_ci_logical_datacenter]

The following attributes in the IIS Virtual Directory [cmdb_ci_iisdirectory] table are populated by collected data:

Attribute label	Attribute name
Alias	alias
Installation directory	install_directory
Name	name

Relationship created for IIS Virtual Directory

Parent class	Relationship type	Child class
IIS Virtual Directory [cmdb_ci_iisdirectory]	Runs on::Runs	Hardware [cmdb_ci_hardware]

The following attributes in the Image [cmdb_ci_os_template] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id

Relationship created for Image

Parent class	Relationship type	Child class
Image [cmdb_ci_os_template]	Hosted on::Hosts	Logical Datacenter [cmdb_ci_logical_datacenter]

The following attributes in the IP Address [cmdb_ci_ip_address] table are populated by collected data:

Attribute label	Attribute name
IP Address	ip_address
Netmask	netmask
IP version	ip_version
Nic	nic

Relationship created for IP Address

Parent class	Relationship type	Child class
IP Address [cmdb_ci_ip_address]	Reference	Network Adapter [cmdb_ci_network_adapter]

The following attributes in the Logical Datacenter [cmdb_ci_logical_datacenter] table are populated by collected data:

Attribute label	Attribute name
Name	name
Region	region
Class	sys_class_name

Relationships created for Logical Datacenter

Parent class	Relationship type	Child class
Logical Datacenter [cmdb_ci_logical_datacenter]	Contains::Contained by	Availability Zone [cmdb_ci_availability_zone]
Logical Datacenter [cmdb_ci_logical_datacenter]	Hosted on::Hosts	Cloud Service Account [cmdb_ci_cloud_service_account]

The following attributes in the Microsoft IIS Web Server [cmdb_ci_microsoft_iis_web_server] table are populated by collected data:

Attribute label	Attribute name
Name	name
Running process command	running_process_command

Attribute label	Attribute name
Version	version
Operational status	operational_status
PID	pid
Install Status	install_status
Type	type

Relationships created for Microsoft iis Web Server

Parent class	Relationship type	Child class
Microsoft iis Web Server [cmdb_ci_microsoft_iis_web_server]	Contains::Contained by	IIS Virtual Directory [cmdb_ci_iisdirectory]
Microsoft iis Web Server [cmdb_ci_microsoft_iis_web_server]	Runs on::Runs	Hardware [cmdb_ci_hardware]

The following attributes in the MS SQL DataBase [cmdb_ci_db_mssql_database] table are populated by collected data:

Attribute label	Attribute name
Data Base	database
Name	name

Relationship created for MS SQL DataBase

Parent class	Relationship type	Child class
MS SQL DataBase [cmdb_ci_db_mssql_database]	Runs on::Runs	Hardware [cmdb_ci_hardware]

The following attributes in the MSFT SQL Instance [cmdb_ci_db_mssql_instance] table are populated by collected data:

Attribute label	Attribute name
Edition	edition
Instance Name	instance_name
Name	name
Operational status	operational_status
Service pack	service_pack
Install Status	install_status

Relationships created for MSFT SQL Instance

Parent class	Relationship type	Child class
MSFT SQL Instance [cmdb_ci_db_mssql_instance]	Runs on::Runs	Hardware [cmdb_ci_hardware]
MSFT SQL Instance [cmdb_ci_db_mssql_instance]	Contains::Contained by	MS SQL DataBase [cmdb_ci_db_mssql_database]

The following attributes in the Network Adapter [cmdb_ci_network_adapter] table are populated by collected data:

Attribute label	Attribute name
DHCP Enabled	dhcp_enabled
Netmask	netmask
Configuration Item	cmdb_ci
Mac manufacturer	mac_manufacturer

Attribute label	Attribute name
MAC Address	mac_address
Name	name

Relationship created for Network Adapter

Parent class	Relationship type	Child class
Network Adapter [cmdb_ci_network_adapter]	Reference	Hardware [cmdb_ci_hardware]

The following attributes in the Serial Number [cmdb_serial_number] table are populated by collected data:

Attribute label	Attribute name
Serial Number	serial_number
Serial Number Type	serial_number_type
Valid	valid

Relationship created for Serial Number

Parent class	Relationship type	Child class
Serial Number [cmdb_serial_number]	Reference	Hardware [cmdb_ci_hardware]

The following attributes in the Software [cmdb_ci_spkg] table are populated by collected data:

Attribute label	Attribute name
Key	key
Name	name
Version	version

Attribute label	Attribute name
Manufacturer	manufacturer

Relationship created for Software

Parent class	Relationship type	Child class
Software [cmdb_ci_spkg]	Reference	Software Instance [cmdb_software_instance]

The following attributes in the Software Instance [cmdb_software_instance] table are populated by collected data:

Attribute label	Attribute name
Install date	install_date
Installed on	installed_on
Name	name

Relationships created for Software Instance

Parent class	Relationship type	Child class
Software Instance [cmdb_software_instance]	Reference	Computer [cmdb_ci_computer]
Software Instance [cmdb_software_instance]	Reference	Hardware [cmdb_ci_hardware]

The following attributes in the Storage Volume [cmdb_ci_storage_volume] table are populated by collected data:

Attribute label	Attribute name
Object ID	object_id

Attribute label	Attribute name
Volume ID	volume_id
Name	name
Size bytes	size_bytes

Relationship created for Storage Volume

Parent class	Relationship type	Child class
Storage Volume [cmdb_ci_storage_volume]	Hosted on::Hosts	Logical Datacenter [cmdb_ci_logical_datacenter]

The following attributes in the Virtual Machine Instance [cmdb_ci_vm_instance] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id
IP Address	ip_address
State	state

Relationships created for Virtual Machine Instance

Parent class	Relationship type	Child class
Virtual Machine Instance [cmdb_ci_vm_instance]	Hosted on::Hosts	Logical Datacenter [cmdb_ci_logical_datacenter]
Virtual Machine Instance [cmdb_ci_vm_instance]	Provisioned From::Provisioned	Image [cmdb_ci_os_template]

Parent class	Relationship type	Child class
Virtual Machine Instance [cmdb_ci_vm_instance]	Provisioned From::Provisioned	Hardware Type [cmdb_ci_compute_template]
Virtual Machine Instance [cmdb_ci_vm_instance]	Use End Point To::Use End Point From	Storage Volume [cmdb_ci_storage_volume]

Related concepts

- [Service Graph Connector for SolarWinds](#)

Accessing the connection details of Service Graph Connector for SolarWinds

You can access the connection details of the Service Graph Connector for SolarWinds in a single view using the common connection framework (CCF) included within the Integration Commons for CMDB (sn_cmdb_int_util) store app.

With the CCF, you can access all the connections used by the Service Graph Connector for SolarWinds. The connection details include the connection alias, connection properties, data sources, and scheduled data imports associated with a connection. You can also test the connection. For more information, see [Accessing the connection details of Service Graph Connectors](#).

Access the details of a SolarWinds connection

Access the details of a SolarWinds connection configured for the Service Graph Connector for SolarWinds.

Before you begin

Role required: admin

Procedure

1. Navigate to **All > Service Graph Connectors > SolarWinds > Connections**.
2. From the **Name** column of the Service Graph Connections list, select a connection.
3. On the Service Graph Connections page, view the connection details such as the connection name, alias, and data sources.
4. Select a related list to view further details of the connection.
5. (Optional) Select the **Test Connection** related link to test the connection.

Configuring air gap connections for Service Graph Connector for SolarWinds

Air gap connections for the Service Graph Connector for SolarWinds enable safe and controlled data transfer without compromising security.

An air gap is a security measure that involves isolating a computer or network and preventing it from establishing an external connection. Air-gapped networks keep critical systems secure by isolating them from external connections to protect the systems from cyber threats. Air gap connections for the Service Graph Connector for SolarWinds facilitate controlled data exchange between the air-gapped networks and external systems while maintaining the integrity of the air-gapped environment.

Air gap connections are available from version 2.5.0 of the Service Graph Connector for SolarWinds. You can add either regular data sources or air gap data sources during the guided setup. To access the guided setup, see [Configure Service Graph Connector for SolarWinds](#).

Perform the following sequence of tasks to configure the air gap solution before you add air gap data sources during the guided setup:

1. [Configure air gap solution for Service Graph Connector for SolarWinds in a high-secure server](#)

2. Configure air gap solution for Service Graph Connector for SolarWinds in a low-secure server
3. Configure air gap solution for Service Graph Connector for SolarWinds in a ServiceNow instance
4. Perform a test data load for the air gap solution for Service Graph Connector for SolarWinds
5. Perform a full data load for the air gap solution for Service Graph Connector for SolarWinds

Configure air gap solution for Service Graph Connector for SolarWinds in a high-secure server

Configure the air gap solution for the Service Graph Connector for SolarWinds in your high-secure server.

Before you begin

Role required: admin

Procedure

1. Navigate to **All > Service Graph Connectors > SolarWinds > Setup > Add Multiple Instances > Add Another Connection**.
2. Download the SgSolarWindsHighSecureSetup.ps1 and GetSolarWindsData.ps1 PowerShell scripts from the link provided in the Add Another Connection section.
3. Log on to your high-secure server.
4. Create a SolarWinds directory.

All the configurations required for the air gap solution are created in the SolarWinds directory.

5. Copy the GetSolarWindsData.ps1 and SgSolarWindsHighSecureSetup.ps1 PowerShell scripts that you downloaded in step 2 to the SolarWinds directory.

6. Start a PowerShell session, and run the `SgSolarWindsHighSecureSetup.ps1` script to generate the JSON configuration file that is used by the `GetSolarWindsData.ps1` script.
 - a. At the prompt, enter the path for the SolarWinds directory that you created in step 4.
 - b. Enter a name for the configuration file, or press the Return key to accept the default value.
 - c. Enter the values for the global properties, or press the Return key to accept the default values.
 - `npmInstalled`: Set the value to `true` if the module is installed on the SolarWinds instance. Set the value to `false` if the module isn't installed on the SolarWinds instance.
 - `samlInstalled`: Set the value to `true` if the module is installed on the SolarWinds instance. Set the value to `false` if the module isn't installed on the SolarWinds instance.
 - `pageSize`: Enter the number of records to be fetched in a SolarWinds API call.
 - `endpoint`: Update the **Host** field with the host name for your SolarWinds instance.
 - `port`: Specify the port on which the SolarWinds instance allows API calls.
 - `credentialsPath`: Specify the location to store the SolarWinds credentials XML file.
 - `logDirectory`: Specify the location to store the log file from the `GetSolarWindsData.ps1` script.
 - d. Specify names for the data source data directories, or press the Return key to accept the default values.

The directories are created if they don't already exist.
 - e. Enter the SolarWinds credentials.

Note: You can either use Windows credential manager to store the user name and password, or implement your own mechanism for storing the credentials. The GetSolarWindsData.ps1 script must have access to the credentials to trigger REST calls.

- f. Exit the PowerShell session.
7. Provide write and delete permissions for the SolarWinds directory and the child directories.

You must provide delete permissions for old records to be cleaned up.

Related tasks

- [Configure air gap solution for Service Graph Connector for SolarWinds in a low-secure server](#)

Configure air gap solution for Service Graph Connector for SolarWinds in a low-secure server

Configure the air gap solution for the Service Graph Connector for SolarWinds in your low-secure server after you finish the configuration in your high-secure server.

Before you begin

Role required: admin

Procedure

1. Navigate to **All > Service Graph Connectors > SolarWinds > Setup > Add Multiple Instances > Add Another Connection**.
2. Download the SgSolarWindsLowSecureSetup.ps1 PowerShell script from the link provided in the Add Another Connection section.
3. Log on to your low-secure server.
4. Create a SolarWinds directory.

All the configurations required for the air gap solution are created in the SolarWinds directory.

5. Copy the SgSolarWindsLowSecureSetup.ps1 PowerShell script that you downloaded in step 2 to the SolarWinds directory.
6. Start a PowerShell session, and run the SgSolarWindsLowSecureSetup.ps1 script.
 - a. At the prompt, enter the path for the SolarWinds directory that you created in step 4.
 - b. Specify names for the data source data directories, or press the Return key to accept the default values.

The directories are created if they don't already exist.
 - c. Exit the PowerShell session.
7. Provide read and delete permissions for the SolarWinds directory.

Related tasks

- [Configure air gap solution for Service Graph Connector for SolarWinds in a ServiceNow instance](#)

Configure air gap solution for Service Graph Connector for SolarWinds in a ServiceNow instance

Configure the air gap solution for the Service Graph Connector for SolarWinds in your ServiceNow instance after you finish the configuration in your high-secure and low-secure servers.

Before you begin

Role required: admin

Procedure

1. Navigate to **All > System Definition > Plugins**, and download the ServiceNow IntegrationHub Action Step – PowerShell plugin.

2. Navigate to **All > Service Graph Connectors > SolarWinds > Setup > Add Multiple Instances > Add Another Connection**.
3. Select the **Add Another Connection** task, and then select **Configure**.
4. In the Select Data Source Type section, select **Airgap Data Sources**, and then select **Add Connection**.
5. Fill in the fields on the Create Connection form.

Create Connection form

Field	Description
Connection Name	Display name for the connection.
MID Server	Name of the MID Server that you deployed in your low-secure server. You can obtain the sys_id of the MID Server from the ecc_agent table.
Parent Directory	Path of the SolarWinds directory where the data is stored in your low-secure server. See Configure air gap solution for Service Graph Connector for SolarWinds in a low-secure server.
Archive Data After Retrieval	Option to archive data after retrieval.
Archive Path [Optional]	Folder where the data is to be archived if the Archive Data After Retrieval check box is selected.

6. Select **Create Connection**.

Related tasks

- [Perform a test data load for the air gap solution for Service Graph Connector for SolarWinds](#)

Perform a test data load for the air gap solution for Service Graph Connector for SolarWinds

After you configure the air gap solution for the Service Graph Connector for SolarWinds, test whether the SolarWinds endpoint is accessible from the high-secure server.

Before you begin

Role required: admin

Procedure

1. Navigate to the SolarWinds directory in your high-secure server.
2. Run the `.\GetSolarWindsData.ps1 -configFilePath CONFIG_PATH` script, where CONFIG_PATH is the location of the JSON configuration file that is generated by the setup script.
The script accesses the SolarWinds Hardware API, and creates a sample file in the Hardware directory and a log file in the Logs directory.

- If the test data load is successful, the log file contains the Test Load status SUCCESS! message.
- If the test data load fails, the log file contains the Error occurred while processing hardware StatusCode 403 message.

If the test data load fails, verify the SolarWinds user credentials and ensure that the user has the required permissions to pull the data from SolarWinds. If the credentials need to be updated, you must run the `SgSolarWindsHighSecureSetup.ps1` PowerShell script again. See [Configure air gap solution for Service Graph Connector for SolarWinds in a high-secure server](#).

Related tasks

- [Perform a full data load for the air gap solution for Service Graph Connector for SolarWinds](#)

Perform a full data load for the air gap solution for Service Graph Connector for SolarWinds

After you verify that the test data load for the air gap solution for the Service Graph Connector for SolarWinds is successful, you can perform a full data load.

Before you begin

Role required: admin

Procedure

1. Navigate to the SolarWinds directory in your high-secure server.
2. Run the `.\GetSolarWindsdata.ps1 -completeLoad $true -configFilePath CONFIG_PATH` script, where CONFIG_PATH is the location of the JSON configuration file that is generated by the setup script.

By default, the existing files in the data source data directories are deleted each time the `.\GetSolarWindsdata.ps1 -completeLoad $true -configFilePath CONFIG_PATH` script is run. You can add the `-clean $false` parameter to the script if you want to retain the existing files.

The script fetches data from all the SolarWinds APIs, and stores the data in the respective directories.

3. After the data is loaded in your high-secure server, transfer the data to your low-secure server using a method of your choice, such as SFTP.
4. Verify that the data is loaded in your low-secure server.
5. Set the full data load job (using the `.\GetSolarWindsdata.ps1 -completeLoad $true -configFilePath CONFIG_PATH` script) to run on

a schedule based on that of the Scheduled Import Job on your Service Graph Connector for SolarWinds instance.

6. Set the data transfer job (from your high-secure server to your low-secure server) on a schedule based on that of the Scheduled Import Job on your Service Graph Connector for SolarWinds instance.
7. Test the air gap connection in your ServiceNow instance.
 - a. Navigate to **All > Service Graph Connectors > SolarWinds > Connections**.
 - b. Select the air gap connection that you created.
 - c. Select the **Test Connection** related link.
When the connection is successful, the **Status** field is set to **Success**. If the connection fails, follow the suggestions for remediation.